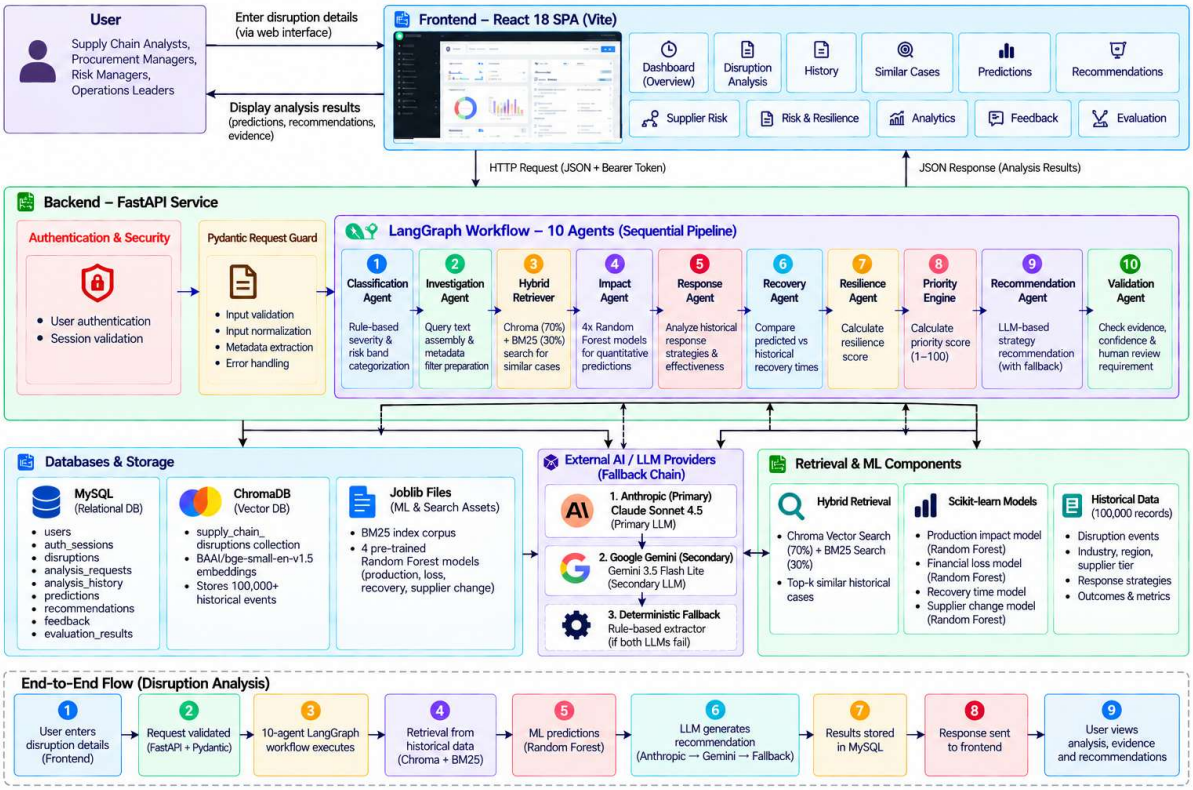

User
Supply Chain Analysts, Procurement Managers, Risk Managers, Operations Leaders
Enter disruption details (via web interface)
Display analysis results (predictions, recommendations, evidence)
Frontend – React 18 SPA (Vite)
Dashboard (Overview)
Disruption Analysis
History
Similar Cases
Predictions
Recommendations
Supplier Risk
Risk & Resilience
Analytics
Feedback
Evaluation
HTTP Request (JSON + Bearer Token)
JSON Response (Analysis Results)
Backend – FastAPI Service
Authentication & Security
User authentication
Session validation
Pydantic Request Guard
Input validation
Input normalization
Metadata extraction
Error handling
LangGraph Workflow – 10 Agents (Sequential Pipeline)
1 Classification Agent
Rule-based severity & risk band categorization
2 Investigation Agent
Query text assembly & metadata filter preparation
3 Hybrid Retriever
Chroma (70%) + BM25 (30%) search for similar cases
4 Impact Agent
4x Random Forest models for quantitative predictions
5 Response Agent
Analyze historical response strategies & effectiveness
6 Recovery Agent
Compare predicted vs historical recovery times
7 Resilience Agent
Calculate resilience score
8 Priority Engine
Calculate priority score (1–100)
9 Recommendation Agent
LLM-based strategy recommendation (with fallback)
10 Validation Agent
Check evidence, confidence & human review requirement
Databases & Storage
MySQL (Relational DB)
users
auth_sessions
disruptions
analysis_requests
analysis_history
predictions
recommendations
feedback
evaluation_results
ChromaDB (Vector DB)
supply_chain_disruptions collection
BAAI/bge-small-en-v1.5 embeddings
Stores 100,000+ historical events
Joblib Files (ML & Search Assets)
BM25 index corpus
4 pre-trained Random Forest models (production, loss, recovery, supplier change)
External AI / LLM Providers (Fallback Chain)
1. Anthropic (Primary) Claude Sonnet 4.5 (Primary LLM)
2. Google Gemini (Secondary) Gemini 3.5 Flash Lite (Secondary LLM)
3. Deterministic Fallback Rule-based extractor (if both LLMs fail)
Retrieval & ML Components
Hybrid Retrieval
Chroma Vector Search (70%) + BM25 Search (30%)
Top-k similar historical cases
Scikit-learn Models
Production impact model (Random Forest)
Financial loss model (Random Forest)
Recovery time model
Supplier change model (Random Forest)
Historical Data (100,000 records)
Disruption events
Industry, region, supplier tier
Response strategies
Outcomes & metrics

End-to-End Flow (Disruption Analysis)
1 User enters disruption details (Frontend)
2 Request validated (FastAPI + Pydantic)
3 10-agent LangGraph workflow executes
4 Retrieval from historical data (Chroma + BM25)
5 ML predictions (Random Forest)
6 LLM generates recommendation (Anthropic → Gemini → Fallback)
7 Results stored in MySQL
8 Response sent to frontend
9 User views analysis, evidence and recommendations